

GUIDING ROBOTIC ENDOSCOPE VIA DEEP LEARNING BASED SURGICAL INSTRUMENT DETECTION

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ABSTRACT

GUIDING ROBOTIC ENDOSCOPE VIA DEEP LEARNING BASED SURGICAL INSTRUMENT DETECTION

Although some minimally invasive surgery procedures are handled by teleoperating a robot from a console, there are still some endoscopic procedures handled by surgeons. These surgeons need to handle the endoscope throughout the surgery whilst they can only operate just another surgical tool resulting in inefficient surgery procedures. The proposed approach enables the surgeon to focus entirely on the surgical procedure while a robotic arm autonomously performs the endoscope positioning task. The system is based on segmenting surgical tools and localizing their tips. We developed two pipelines — one performs morphological image processing operations to localize tool-tips after tool segmentation, the other employs deep learning architectures for concurrent segmentation and tool-tip detection. Three architectural configurations were systematically evaluated: U-Net as the baseline, ART-Net as the primary segmentation backbone, and ART-Net augmented with Convolutional Block Attention Module (CBAM) layers to investigate the effect of channel and spatial attention on both segmentation and localization performance. Experimental results indicate that ART-Net substantially outperforms U-Net in segmentation across all dataset configurations, while CBAM integration introduces a precision-recall trade-off whose benefit is contingent on the geometric properties of the instruments present in the target domain. In the tool-tip localization stage, the heatmap-based deep learning approach yields superior spatial precision compared to the morphological pipeline.

ÖZET

DERİN ÖĞRENME TABANLI CERRAHİ ALET TESPİTİ YOLUYLA ROBOTİK ENDOSKOPUN YÖNLENDİRİLMESİ

Bir kısım minimal invazif cerrahi operasyonlarda endoskop yönlendirmesi her ne kadar konsol üzerinden gerçekleşse de, endoskopik ameliyatlara hala cerrahlar tarafından yönlendirilmektedir. Bu şekilde yönlendirme, cerrahlar açısından verimsiz olup ameliyat boyunca yalnızca tek bir cerrahi alet kullanabilmektedirler. Önerilen yaklaşım cerrahların tamamen ameliyat aletlerine odaklanmasını sağlarken, endoskop yönlendirilmesi bir robotik kol üzerinden otonom bir şekilde gerçekleştirebilmektedir. Yaklaşım sahnede var olan aletlerin bölütlenmesi ve konumlandırılmasından ibarettir. İki farklı işlem hattı geliştirdik: Birinci yöntem alet bölütlemesi sonrası morfolojik operasyonlar uygulayarak alet ucu tespit ederken ikinci yöntem alet bölütlemesi ve alet ucu tespiti için derin öğrenme mimarileri kullanmaktadır. 3 farklı mimari konfigürasyon, U-Net, ART-Net ve ara katmanlara CBAM eklenmiş ART-Net, sistematik bir şekilde değerlendirildi. Deneysel sonuçlar, ART-Net tabanlı mimarilerin tüm veri seti yapılandırmalarında segmentasyon görevinde U-Net'i önemli ölçüde geride bırakırken, CBAM entegrasyonu bir hassasiyet-geri çağırma dengesi (precision-recall trade-off) ortaya çıkarmakta olup bu dengenin faydası, hedef sahnedeki aletlerin geometrik özelliklerine bağlıdır. Alet ucu konumlandırma aşamasında ise, ısı haritası tabanlı derin öğrenme yaklaşımı morfolojik konumlandırma yöntemine göre daha hassas koordinatlar tahmin edebilmektedir.

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CHAPTER 1

INTRODUCTION

Minimally Invasive Surgery is a type of surgical procedure in which, rather than fully opening the surgical region, only small incisions are made to provide access for surgical instruments and an endoscopic camera. These procedures offer significant advantages for patients, including reduced recovery time and minimal postoperative scarring. Despite these benefits, minimally invasive surgeries introduce considerable challenges for surgeons. In addition to the high precision required during the operation, the loss of direct visual access to the patient's anatomy and the presence of occlusions caused by blood, tissue, and surgical instruments compel surgeons to rely heavily on the endoscopic camera view. Consequently, surgeons are required to frequently shift their attention between the surgical instruments and the endoscopic camera, which may lead to distraction and increased fatigue during the procedure.

A common approach to alleviate this issue is to assign an additional surgeon or assistant solely responsible for manipulating the endoscopic camera, allowing the primary surgeon to focus entirely on the surgical procedure. Although this approach may initially appear practical and beneficial, it results in inefficient utilization of hospital resources, since the secondary surgeon is dedicated solely to camera control rather than participating in other surgical procedures (Stott et al. 2017). Furthermore, communication mismatches between the primary surgeon and the camera operator, together with distractions, fatigue, and involuntary movements of the camera manipulator, may increase the overall duration of the surgery. To overcome the limitations associated with human-operated camera control, passive or active endoscope holder systems can be employed.

Various interfaces for controlling the endoscope have been employed, including joystick-based systems (Kim, Park, and Lee 2019), voice-command interfaces (Chakradeo et al. 2006), eye-tracking methods (Fujii et al. 2017), and inertial measurement unit (IMU)-based motion-capture approaches (M. Dede et al. 2020). The obvious problem when using a joystick is that the surgeon needs to release one of the surgical tools to steer the endoscope. General problems when using voice commands are the noisy surgery

room and the extra mental efforts of the surgeon to calibrate his voice and commands with respect to the voice recognition algorithm. Tracking the eyes of the surgeon brings new challenges such as keeping the surgery room at the same brightness level. Although motion capturing system with an IMU-based sensory system may seem like an intuitive way to control the robot, the surgical tool holding style of the surgeon changes from one tool to another and hence, the motion capabilities of the surgeon’s hand vary vastly during the surgery.

Despite advances in robotic and laparoscopic surgery, endoscope guidance remains predominantly dependent on manual operation or non-semantic control interfaces that lack any understanding of the surgical scene. Existing approaches — whether joystick-based, voice-commanded, or motion-captured — treat the endoscope as an isolated instrument rather than as an active observer embedded within a semantically rich environment. Although prior work in computer vision has explored surgical instrument detection as an isolated perception task, these efforts have rarely been connected to a downstream automation objective, leaving the perception-to-action loop largely unclosed.

The endoscopic video stream itself, however, continuously captures the surgical scene and inherently encodes the spatial and contextual information necessary for intelligent camera control. Surgical instruments — as the primary objects of interest during a procedure — serve as natural anchors for scene understanding and endoscope guidance. Detecting and localizing these instruments in real time therefore constitutes the critical technical sub-problem underlying endoscope automation. This task is non-trivial: laparoscopic scenes present severe visual challenges including partial and full occlusion by tissue and blood, specular reflections on metallic instrument surfaces, significant variation in instrument scale and orientation, and substantial domain shift across different surgical settings and camera systems.

This thesis addresses the problem of AI-driven surgical endoscope automation through the lens of instrument detection, exploring how reliable real-time perception of surgical instruments in laparoscopic video can serve as the foundation for an intelligent, autonomous endoscope guidance system.

The tool tip localization pipeline developed in this thesis operates as a core component of the CleverScope system — a TÜBİTAK-funded surgical robotics project developed at the Izmir Institute of Technology Robotics Laboratory (C. Dede et al. 2023). The

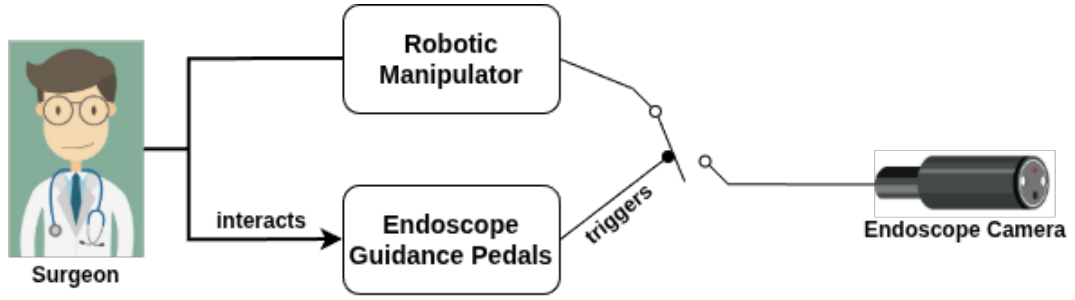


Figure 1.1. Automated guidance by the developed algorithms is activated while the foot pedals are kept pressed.

CleverScope system aims to relieve the surgeon from manually holding and directing the endoscope during endonasal procedures, instead delegating this task to an industrial collaborative robot equipped with a custom-designed end-effector. This allows the surgeon to operate both hands freely on other surgical instruments while the robot maintains and adjusts the endoscope’s field of view autonomously.

At each frame, the endoscope captures the live surgical scene, which is fed into the localization pipeline to detect the position of the surgical tool tip within the image. Based on the detected position’s deviation from the image center, the system computes a set of control outputs — namely, the movement angle, zoom level, and motion speed — which are relayed as driving commands to the robot. The surgeon interacts with the system via two foot pedals: the first triggers the pan movement of the endoscope, while the second controls zoom in and out which is shown in Figure 1.1. Although the tool tip localization pipeline developed in this thesis serves as the perceptual backbone of the CleverScope system, the computation of robot motion commands and the control architecture are outside the scope of this work.

1.1. Contributions of the thesis

To summarize the contribution of this work:

(1) We introduce a deep-learning based tool tip localization method and compare it with the more conventional approach where morphological image processing techniques are employed to find the tool tips after an initial segmentation step (Gruijthuijsen et al. 2022).

(2) Although there are previous examples of deep-learning based tool tip detection

approaches, there are limited to the scenes containing one tool (Hasan et al. 2021) or the tools have significantly different colored tool tips (C. Zhang et al. 2023). To the best of our knowledge, we are the first to develop a deep learning based tool tip detection system that handles multiple tools without tool tip markers.

(3) We introduce a novel approach to set the zoom value of the endoscope dynamically depending on the area covered by the surgical tools.

(4) In addition to the surgical instrument datasets found in the literature, two separate multi-instrument markerless endoscopic surgery datasets were collected, labeled, and made publicly available.

1.2. Organization of the thesis

The remainder of this thesis is organized as follows. Chapter 2 reviews existing marker-based and marker-less approaches for surgical instrument detection and localization, as well as prior work on automated endoscope motion control. Chapter 3 describes the proposed vision-based methodology for endoscope guidance, including the data collection procedure and dataset. Chapter 4 presents the experimental setup, evaluation metrics, and quantitative results. Chapter 5 concludes the thesis with a discussion of the findings and directions for future work.

CHAPTER 2

RELATED WORK

2.1. Marker-based Instrument Localization

To enable autonomous endoscope guidance in laparoscopic procedures, several studies have proposed the attachment of artificial markers to surgical tools. Different types of markers have been developed to improve instrument detectability under varying occlusion conditions commonly encountered during surgery.

Early approaches relied on modifying the appearance of instruments, such as applying distinct color coatings (Ko et al. 2005; Tonet et al. 2007), which contrast with the surgical background and allow detection through simple thresholding operations (Bouarfa et al. 2012). Other methods used optical trackers that emit non-visible wavelengths of light (Speidel et al. 2006), or acoustic tracking systems that estimate the three-dimensional coordinates of transmitters attached to the instruments (Chmarra, Grimbergen, and Dankelman 2007).

Although these approaches simplify the detection process by reducing the complexity of visual scene analysis, they introduce practical challenges related to sterilization, modification, and installation of the instruments. Markers must be carefully positioned to remain visible to the endoscope, and adjustments are often required before each procedure (Bouget et al. 2016). In addition, occlusions caused by blood, tissue, or surgical materials can block markers, resulting in tracking failures and reduced system reliability.

2.2. Markerless Instrument Localization

Markerless localization of surgical instruments primarily leverages the intrinsic visual features within the surgical field. These features are extracted from the instruments' geometric profiles and their spatial orientations relative to the imaging sensor. Given that clinical workflows require a clear view of the operative site, both the instruments and the anatomical region of interest typically remain within the camera's field of view. To achieve

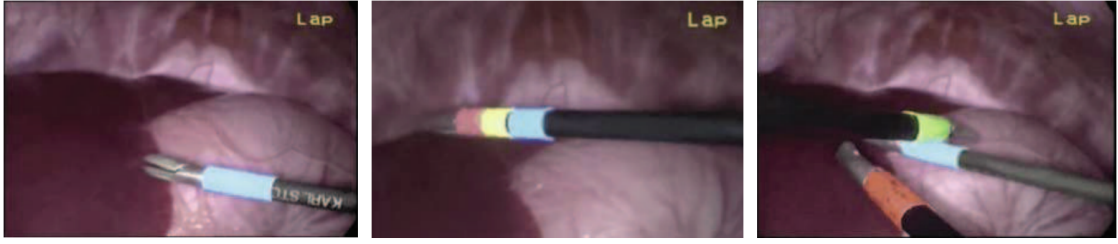


Figure 2.1. Example for Marker-based Surgical Instrument Localization (Source: (Bouarfa et al. 2012))

marker-free localization, researchers have increasingly deployed high-performance object detectors (Iovene et al. 2024), or pixel-wise segmentation frameworks to isolate and identify tool components.

Beyond simple detection, segmentation-based approaches (Cerón et al. 2022; Laina et al. 2017; Kanakatte et al. 2020) provide the pixel-level granularity required for high-fidelity tasks such as instrument tracking and keypoint extraction (Lee et al. 2020; Du et al. 2018). By isolating the tool’s silhouette, these methods identify functional landmarks—such as the tip or joints—to monitor the surgeon’s economy of motion and calculate tool-to-tissue distances. These 2D masks further serve as the foundation for 3D reconstruction, where stereo-matching or depth-estimation models infer the tool’s spatial pose and volume (Hasan et al. 2021; M. Allan et al. 2018). This 3D spatial awareness is critical for maintaining correct depth perception in Augmented Reality (AR) overlays, ensuring that virtual preoperative data is accurately registered relative to the physical instruments.

Attention mechanisms have been incorporated into segmentation architectures to improve their ability to focus on task-relevant regions while suppressing background noise. Among these, the Convolutional Block Attention Module (CBAM) (Woo et al. 2018) applies sequential channel and spatial attention gates to intermediate feature maps, allowing the network to selectively emphasize informative channels and spatial locations. This lightweight yet effective module has been adopted in surgical instrument segmentation to handle the challenges posed by specular reflections, low contrast between instruments and tissue, and cluttered operative backgrounds. CBAM was integrated with a multi-scale feature fusion strategy to achieve real-time instance segmentation of surgical instruments (Cerón et al. 2022), demonstrating that attention-guided feature refinement can preserve fine-grained boundary details under strict latency constraints. Building on this direc-

tion, prior work introduced UniSurg, a unified multitask framework for robotic surgical scene understanding that incorporates CBAM to jointly address instrument segmentation alongside complementary scene understanding tasks (Shen et al. 2026), showing that shared attention-enhanced representations benefit multiple surgical perception objectives simultaneously.

The lack of available data and the difficulty of obtaining and annotating surgical scene images have led researchers to explore generative solutions for synthetic surgical scene generation. Occlusions caused by blood, cloth, or smoke make both data collection and annotation highly challenging. To address these issues, simulation-based surgical scenes containing instruments and real-world images with segmentation labels have been generated using generative adversarial network (GAN)-based models (Pfeiffer et al. 2019; Colleoni and Stoyanov 2021; Colleoni, Edwards, and Stoyanov 2020). It is also possible to bypass the explicit image generation phase and instead apply semi-supervised approaches using unlabeled real images together with annotated simulation images (Sahu et al. 2020).

2.3. Automated Endoscope Motion Based on Instrument Localization

Beyond tool detection itself, recent research has increasingly focused on exploiting localization output to enable autonomous or semi-autonomous control of the endoscopic camera. Accurate estimation of the surgical tool’s position, particularly the tip location, is essential for dynamically adjusting the camera viewpoint to maintain optimal visibility of the operative field. Unlike traditional localization methods that solely identify tool presence, these approaches integrate spatial and temporal cues to inform real-time camera motion. Deep learning-based segmentation and 3D reconstruction techniques have been widely explored for this purpose, offering robustness and adaptability in complex surgical environments. Building on this direction, several studies have proposed pipelines that couple tool tip localization with automatic camera control mechanisms to improve surgical efficiency and reduce the cognitive load on the surgeon.

To detect the tips of surgical instruments, an encoder–decoder-based deep learning segmentation model was proposed (C. Zhang et al. 2023). The dataset was collected under two experimental settings—on a suturing board and on raw pork liver—to simulate different surgical conditions. In that study, the tool tips were assigned distinct colors

to differentiate them from the shaft of the surgical instruments in the dataset. Thus, the segmentation model mainly exploits color differences. Another limitation is that the model does not detect individual tool tips, which makes interpretation based on multiple tips impossible.

An encoder–decoder-based deep learning segmentation model was proposed in (Gruijthuisen et al. 2022) to detect the tips of surgical instruments. Their method employs morphological image processing techniques following a deep learning-based tool segmentation step. In our study, we take this approach as a baseline and compare it with the proposed fully deep learning-based approach, where tool tips are also detected via a secondary localization network head.

Recent literature has also explored diverse tracking frameworks, ranging from the use of geometric and kinematic constraints to ensure identity consistency in laparoscopic environments (Y. Zhang et al. 2026), to the implementation of hybrid tracking modules that combine deep learning with optical flow to reduce latency in exoscope systems (Iovene et al. 2024). While these approaches significantly improve temporal stability and spatial accuracy, their experimental validations were primarily limited to a single instrument type, typically the surgical grasper. Consequently, there remains a critical need for generalizable tracking solutions capable of maintaining performance across broader surgical scenes involving tools with varying geometries and functionalities.

CHAPTER 3

METHODOLOGY

3.1. Description of Automated Endoscope Motion Pipeline

The automated endoscope motion pipeline is designed to analyze surgical endoscopic video frames and extract the position of surgical instruments, with a particular focus on accurate tool tip localization. In parallel to several previous studies (Gruijthuisen et al. 2022; Hasan et al. 2021) the system operates in two main stages: tool segmentation and tool tip detection. The segmentation stage (Section 3.1.1) isolates surgical tools from the background, while the tool tip detection stage (Section 3.1.2) determines the precise coordinates of each tool tip. The output of this pipeline is later used for endoscope motion automation as explained in Section 3.1.3.

3.1.1. Segmentation

To identify the presence and spatial extent of surgical tools, segmentation-based deep neural network models are employed. These models classify each pixel in the frame as either instrument or background, thereby producing a binary mask that delineates the tool regions. When one or more instruments are present, the corresponding masks are extracted to isolate individual tool instances for further analysis. Two different segmentation models were utilized in this study: U-Net (Ronneberger, Fischer, and Brox 2015) and ART-Net (Hasan et al. 2021).

U-Net is a convolutional neural network architecture originally proposed for biomedical image segmentation. Hence, it has been popular among biological and medical domains (Chen et al. 2024; Deininger, Jung-Klawitter, and Mikut 2023; Kahveci et al. 2024; Kahveci et al. 2026). It follows an encoder–decoder design in which the encoder progressively downsamples the input to capture semantic context, while the decoder upsamples feature maps to recover spatial precision. Skip connections between corresponding encoder and decoder layers enable the network to combine low-level spa-

tial information with high-level semantic features, resulting in accurate and fine-grained segmentation even with limited training data. Given its proven effectiveness across a wide range of binary and multi-label segmentation tasks, U-Net was employed as a baseline model in this study. It served both to evaluate the suitability of the dataset for surgical instrument segmentation and to provide a reference point for comparison with more advanced architectures.

ART-Net (Augmented Reality Tool Network) is a deep neural network for joint tool detection, segmentation, and geometric primitive extraction (Hasan et al. 2021). The network features a single encoder–multi-decoder architecture, one for tool detection, one for segmentation, and three regression branches for extracting geometric primitives (edge-lines, mid-line, and tool tip). The encoder is built from five convolutional blocks with 13 layers following the VGG-16 backbone. Each decoder uses depth-wise separable convolutions (inspired by Xception) to make inference lightweight for real-time surgical applications. To preserve spatial accuracy lost during down sampling, ART-Net introduces a Full Resolution feature-map Generator (FrG) skip connection that restores high-resolution details and reduces checkerboard artifacts. The network optimizes a combined cross-entropy loss and intersection over union value (IoU) (Eq. 3.2), which compensates for the severe foreground–background imbalance typical of surgical images.

$$L_{CE} = -\frac{1}{N} \sum_{i=1}^N (y_i \log \hat{y}_i + (1 - y_i) \log (1 - \hat{y}_i)) \quad (3.1)$$

$$L_{seg}(y, \hat{y}) = L_{CE} + 1 - \frac{\sum_{i=1}^N y_i \hat{y}_i}{\sum_{i=1}^N y_i + \sum_{i=1}^N \hat{y}_i - \sum_{i=1}^N y_i \hat{y}_i} \quad (3.2)$$

where N denotes the total number of pixels considered in the loss computation, y_i and $\hat{y}_i$ are the ground truth and the predicted probability of belonging to a surgical tool for each pixel, respectively. Both models serve as the foundation for downstream tool tip localization by providing accurate tool masks that can be post-processed using morphological operations to enhance contour smoothness and reduce noise. Section 4.3.1 compares the segmentation performances of both models with various datasets.

3.1.2. Tool Tip Detection

3.1.2.1. Tool Tip Detection via Morphological Algorithm

After extracting the segmentation mask of the surgical instrument(s) from each image, a series of morphological post-processing operations are applied to enhance the quality of the mask and facilitate accurate tool tip localization. The raw segmentation output obtained from the model often contains small isolated artifacts or incomplete mask regions due to imperfect boundary predictions, which can be seen on Figure 3.1b. To mitigate these issues, morphological opening and closing operations are sequentially applied with a dynamic kernel sizes based on the present tools' thickness. The thickness is calculated by applying distance transform to the binary mask, thresholding pixels with intensity 80% or higher, and finally averaging them and doubling them in order to find the diameter. The opening operation, which consists of an erosion followed by a dilation, removes small noise components and thin protrusions around the outer edges of the instrument masks. Conversely, the closing operation, performed as a dilation followed by an erosion, fills small holes or gaps inside the instrument regions that may have been missed during segmentation. These complementary operations not only improve the structural consistency of the masks but also assist in separating overlapping regions when two or more instruments appear to intersect within the same frame. As a result, a noise-free and topologically consistent binary mask is obtained for each visible surgical instrument that can be seen on Figure 3.1c.

Connected Component Analysis (CCA) is applied to the refined binary mask to systematically identify and spatially isolate discrete foreground regions, each corresponding to an individual surgical instrument present within the scene. Each connected region, referred to as a blob, is extracted along with its associated geometric properties, most notably its area. To suppress spurious detections arising from residual noise or imaging artifacts, a minimum area threshold of 0.5% of the total image area is enforced; any isolated region falling below this threshold is classified as noise and subsequently discarded from further processing.

Following the filtering stage, morphological skeletonization is applied independently to each retained blob, reducing the binary region to a one-pixel-wide medial axis representation. This process, grounded in iterative morphological thinning, preserves the

topological structure of the original region while distilling it into its most compact geometric form. The resulting skeleton effectively encodes the central longitudinal axis of the surgical instrument, capturing its geometric trajectory through the image plane — from the proximal point at which the instrument enters the camera’s field of view to its distal tip. This medial axis representation serves as a robust basis for subsequent geometric reasoning, including shaft orientation estimation and tip localization.

To estimate the tip point of the tool, pixels within the skeleton that have only a single neighbor are identified as skeleton endpoints, as illustrated in Figure 3.1d. Once the endpoints are determined, the Euclidean distance from each endpoint to the image center is computed, and the endpoint with the minimum distance is selected as the tip point. This approach is based on the assumption that the tool tip is consistently positioned closer to the center of the image relative to the proximal end, which typically remains near the image boundary.

This pipeline — comprising morphological refinement, component analysis, and geometric evaluation following a tool segmentation step — can be considered as the classical approach when compared to fully deep learning based approach given below. With experiments, we will compare their performances with respect to each other.

3.1.2.2. Tool Tip Detection via Deep Learning

The ART-Net structures the training process into two distinct stages. In the first stage, the authors trained and evaluated only the segmentation decoder using the EndoVis dataset to establish a baseline for comparison. In the second stage, all decoder branches—including segmentation and tool tip detection—were jointly trained and evaluated on their own custom dataset (Hasan et al. 2021).

In this study, a similar two-stage training strategy was adopted (Figure 3.2). The training procedure was conducted in two stages. In the first stage, the segmentation sub-network of ART-Net was trained to obtain robust feature representations of surgical instruments under diverse conditions. In the second stage, the encoder weights were frozen to retain the learned representations, and the tool tip detection and segmentation sub-networks were jointly trained. Several output formats were explored for tool tip prediction, including (i) direct regression of the x–y coordinates of tool tips and (ii) generation of a binary mask in which each tool tip was represented by a Gaussian-shaped

region with a radius of 10 pixels.

While the coordinate-based approach produced accurate predictions for images containing a single tool, it failed to generalize to multi-tool scenes due to the lack of permutation invariance between tool instances. Consequently, the Gaussian mask representation was selected as the final output format, which is shown in Figure 3.3.

For the loss function, mean squared error (MSE) was employed, where the predicted tool tip masks were compared with the ground truth on a pixel-wise basis and summed across all pixels to compute the total loss. The training dataset consisted exclusively of scenes containing one or more tools with non-articulated bodies, consistent with the assumptions of the proposed pipeline.

Following training, the model produces output heatmaps in which regions corresponding to instrument tips exhibit higher activation intensity relative to surrounding areas, as shown in Figure 3.4b. Since the predicted heatmap values can vary across a wide range, min-max normalization is applied to rescale them to the interval $[0, 1]$, ensuring compatibility with the subsequent post-processing steps. A Gaussian blur is then applied to the normalized heatmap to spatially smooth the activation, consolidating multiple nearby high-intensity regions into a single coherent peak, as demonstrated in Figure 3.4c. A Gaussian-shaped spatial prior, centered at the image center and decaying to zero toward the image boundaries, is incorporated into the smoothed heatmap as a center bias, reflecting the tendency of the tool tip to appear near the image center, as shown in Figure 3.4d. This bias is added only to pixels with values above a threshold of 0.3, at a reduced contribution of 70%, so as not to override strong off-center activations. Its primary role is to resolve ambiguity in cases where two candidate peaks have similar heatmap intensities but differ in their proximity to the image center. Local maxima are then extracted by applying a sliding window maximum filter, in which a pixel is considered a peak candidate if it equals the maximum value within a surrounding neighborhood of size 25×25 pixels and exceeds an absolute intensity floor of 0.3. The surviving candidates are subsequently grouped using connected component labeling, as a single true peak may activate several adjacent pixels at equal intensity. For each resulting cluster, an intensity-weighted centroid is computed to obtain a precise peak coordinate (x, y) .

In cases where candidate points fall outside the predicted instrument regions, a validation step is introduced. Each candidate is evaluated against a slightly dilated version

of the segmentation mask produced by the segmentation branch (Figure 3.4e), where the dilation provides a degree of tolerance for minor spatial misalignment between the two network outputs. Candidates lying outside the dilated mask are discarded. When multiple candidates are associated with the same instrument, the segmentation mask is decomposed into individual instrument regions using connected component analysis, and each candidate is assigned to the region it spatially falls within. If a single region contains multiple candidates, only the one with the highest heatmap intensity is retained (Figure 3.4f).

3.1.3. Endoscope Guidance Based on Tool Tip Detection

Following the extraction of surgical tool tips using the deep learning–based detection pipeline, the angular position of the instruments relative to the image center was computed for each video frame. When a single tool tip was detected, its image coordinates were directly used for angle calculation. When multiple tool tips were detected, the centroid of the detected tips was computed by taking the arithmetic mean of their coordinates. All detected tool tips were considered equally relevant and therefore assigned equal weight in the centroid calculation.

The angular position was defined as the angle between the positive horizontal axis and the line segment connecting the image center to the reference point (i.e., the detected tip or the computed centroid). Angles were measured counterclockwise within the range of $[0^\circ, 360^\circ)$, where 0° corresponds to the leftward horizontal direction, 90° to the upward vertical direction, 180° to the rightward horizontal direction, and 270° to the downward vertical direction (Figure 3.5).

Zoom control was derived from the segmentation output of the model. Because the endoscope employs a monocular camera, direct depth estimation was not available. Therefore, the relative area of the binary surgical instrument mask was used as an indirect indicator of instrument–endoscope distance. Specifically, the percentage of image pixels classified as instrument was computed for each frame.

Two threshold values—denoted as the zoom-in and zoom-out thresholds—were used to determine zoom actions. Initially, these thresholds were empirically selected. However, variability in surgeon preference observed during user testing indicated that

fixed thresholds were not universally optimal. Consequently, the system was modified to allow surgeon-specific threshold selection after a short calibration phase.

If the instrument mask percentage exceeded the zoom-out threshold, a zoom-out signal was generated, indicating that the instruments occupied an excessive portion of the visual field. Conversely, if the percentage fell below the zoom-in threshold, a zoom-in signal was generated. If the percentage remained between the two thresholds, no zoom adjustment was requested. (Figure 3.6)

For each incoming video frame, the computed angular position and zoom command were transmitted to the endoscope controller. These parameters were updated continuously on a frame-by-frame basis. However, physical motion of the endoscope was executed only when the surgeon activated the corresponding foot pedal (for pan motion or zoom). Upon pedal activation, the endoscope moved according to the most recently computed control parameters. Motion continued until the surgeon released the pedal or until a predefined mechanical boundary.

After profiling each stage of the proposed framework, the complete segmentation and morphological tool tip detection pipeline requires approximately 68 ms per frame, corresponding to an average processing speed of about 14 frames per second (FPS). In comparison, the deep learning-based pipeline achieves an average throughput of approximately 10 FPS under the same experimental conditions. These frame rates are sufficient to support near real-time visualization, which is generally considered adequate for screen-guided surgical manipulation.

3.2. Dataset

The performance of the proposed surgical tool segmentation and tip localization pipeline largely depends on the quality and diversity of the dataset used for training and evaluation. To construct a comprehensive dataset for surgical instrument segmentation, multiple public and custom datasets were combined. The primary motivation for creating a diverse dataset was to ensure that the model could accurately distinguish surgical tools across a variety of scenes and anatomical regions. However, many publicly available datasets contain surgical instruments with articulated bodies or a specific tool. To address this, additional data were collected, annotated, and integrated into the training set.

The final dataset comprises images from four distinct sources:

1. **2017 Endovis Robotic Instrument Segmentation Test Dataset** (Max Allan et al. 2019): This publicly available dataset, published on Kaggle, contains 350 high-quality surgical images with a resolution of 1280×1024 pixels. Each image includes 1–3 surgical instruments, and corresponding binary masks are provided. The dataset features relatively clean scenes without occlusions from blood, smoke, or surgical drapes (Figure 3.7a).
2. **Surgical Scene Segmentation in Robotic Gastrectomy Dataset** (Yoon et al. 2022): This dataset consists of 4,510 images with semantic masks for multiple surgical tools and anatomical structures. While primarily used for generating synthetic data to augment segmentation performance, a subset of 2174 images containing only real images of surgical tools was selected for this study. Compared to the Endovis dataset, these images are more challenging due to the presence of occlusions, poor lighting conditions, and occasional out-of-focus frames. The resolution of these images is 1280×1024 pixels (Figure 3.7b).
3. **Hacettepe University Surgery Dataset** : With the help of surgeons from Hacettepe University (Ankara, Turkey), our team collected approximately 1005 images from real surgical procedures. Because the dataset was initially unlabeled, binary masks were manually created for each image. 492 of the images were used for segmentation and tool tip localization training, and the remaining 513 were used in testing the trained models. Each frame contains one or two instruments at a resolution of 1920×1080 pixels. Due to the real surgical environment, these images exhibit variations in lighting, obstructions, and visual noise reflecting realistic operating conditions (Figure 3.7c).
4. **Handcrafted Endoscope Dataset** : To simulate surgical scenes in a controlled environment, our team collected a total of 621 images using an endoscope on a red bell pepper along with surgical instruments. These images were segmented and labeled with instrument tip tags. 464 images were used for segmentation and tool tip training, while 157 images were used for instrument tip performance evaluation. The inside of the pepper was dyed red to mimic human tissue, and the images were acquired at a resolution of 1024×768 pixels (Figure 3.7d).

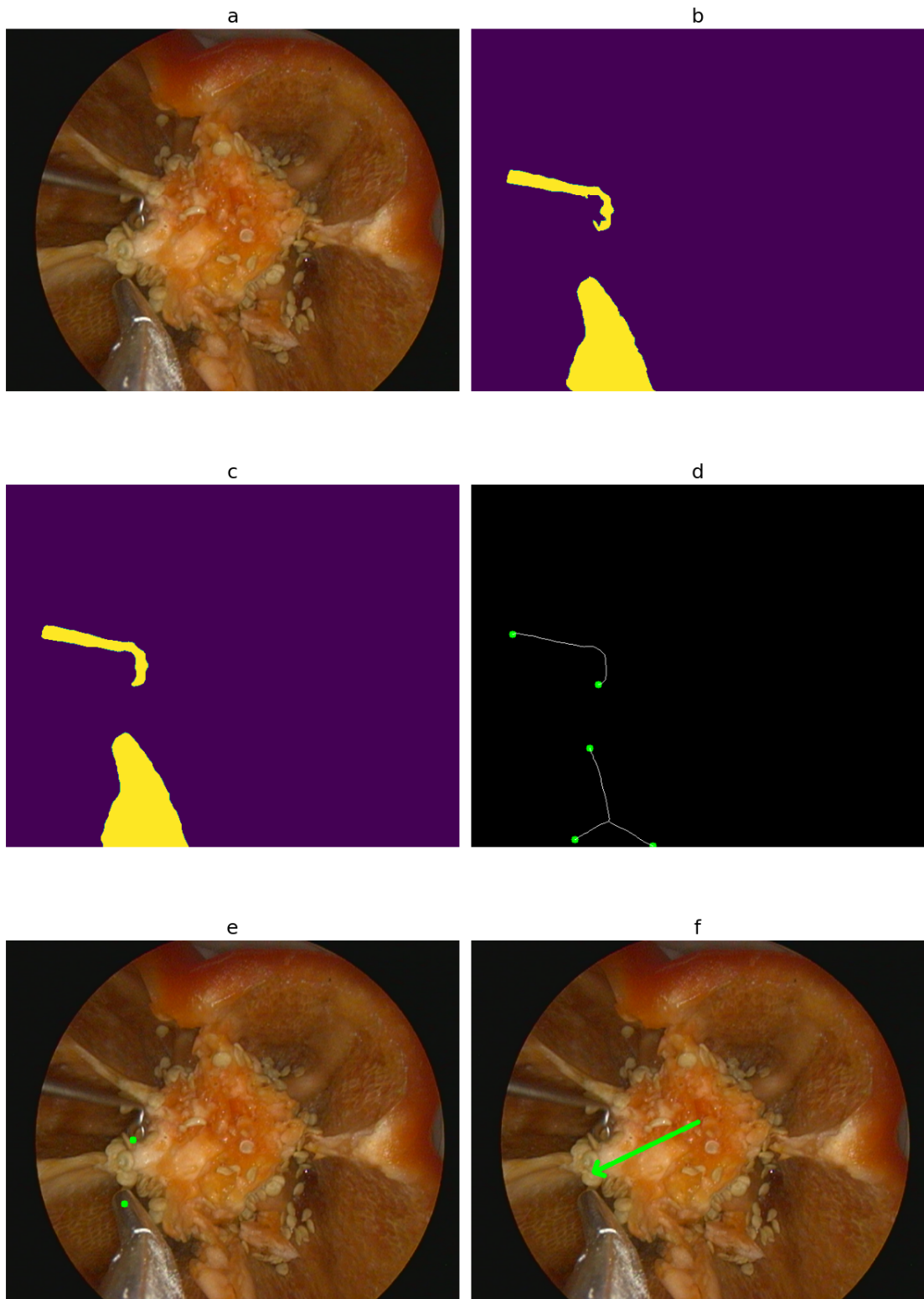


Figure 3.1. Morphological tool tip detection pipeline; a. Raw surgical image, b. Segmentation output, c. Opening & Closing operations applied, d. Masks skeletonized & separated, e. Tips located, f. Mean tip point and angle values calculated.

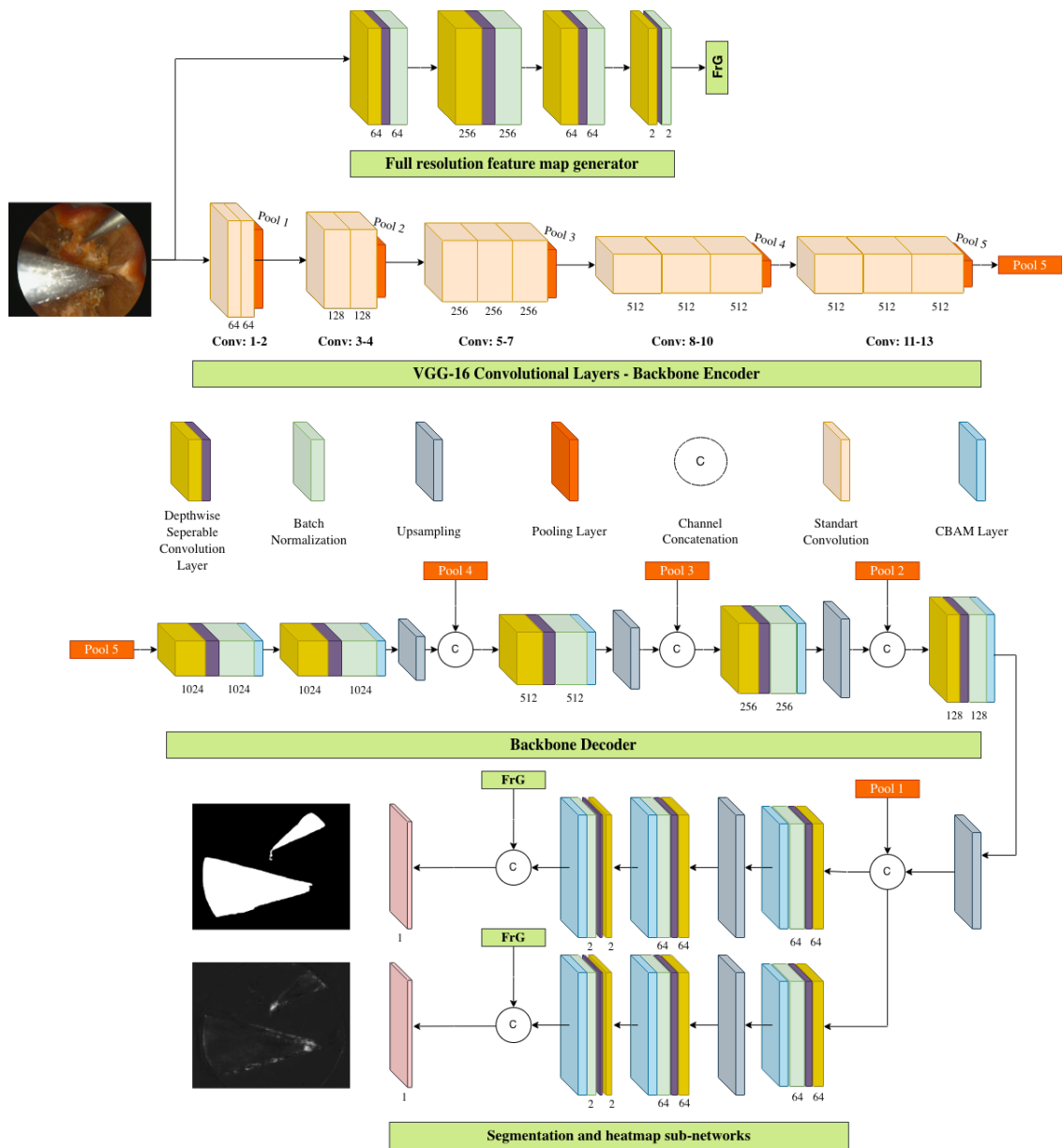


Figure 3.2. Deep Learning based segmentation & tool tip detection model.



Figure 3.3. Left to right: Surgical scene, Segmentation label, tool tip label

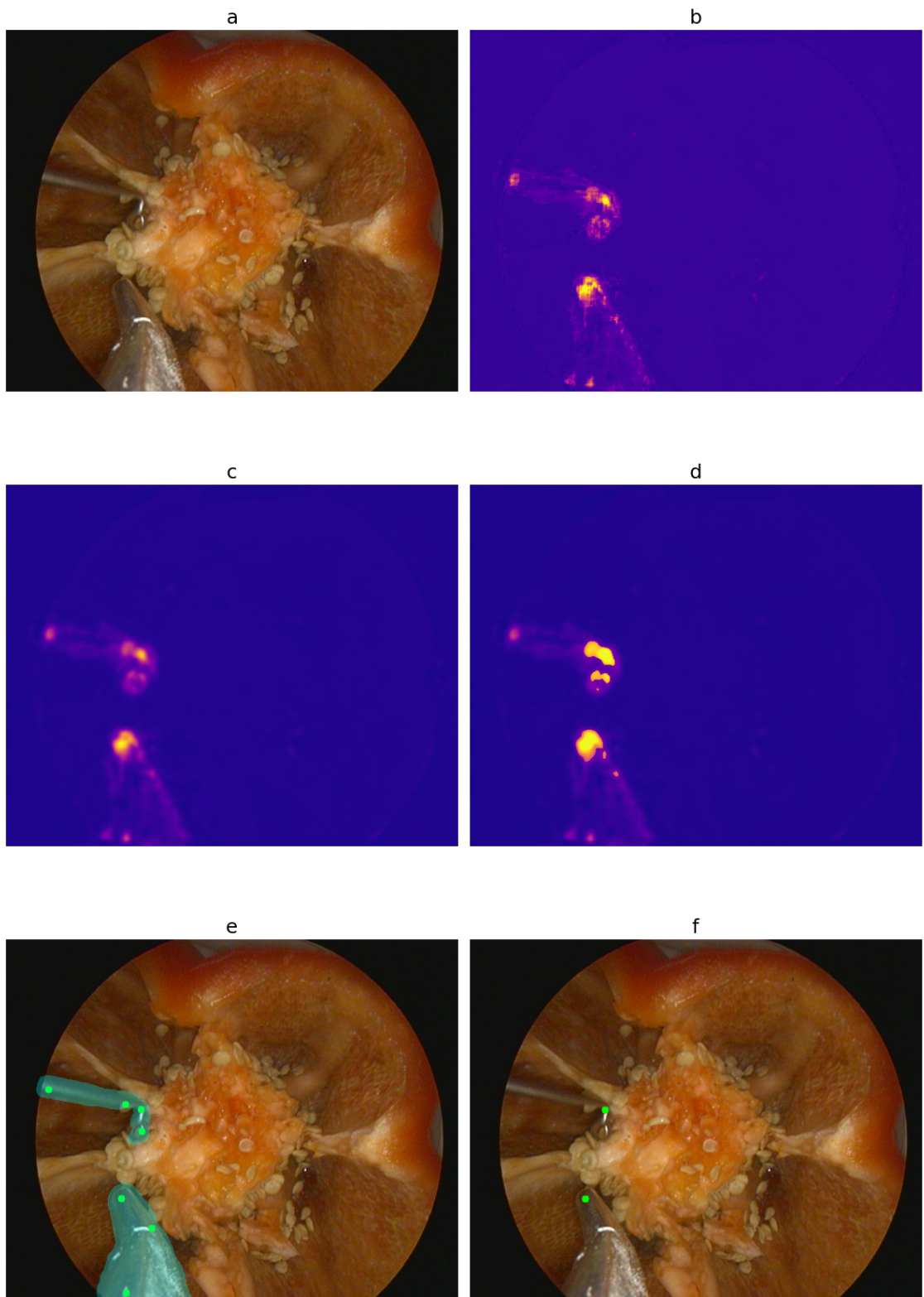


Figure 3.4. Deep Learning based tool tip detection pipeline; a. Raw surgical image, b. Heatmap network output, c. Smoothed Heatmap, d. Smoothed heatmap with center bias, e. Calculated & validated tip point candidates, f. Final tip-points of instruments.

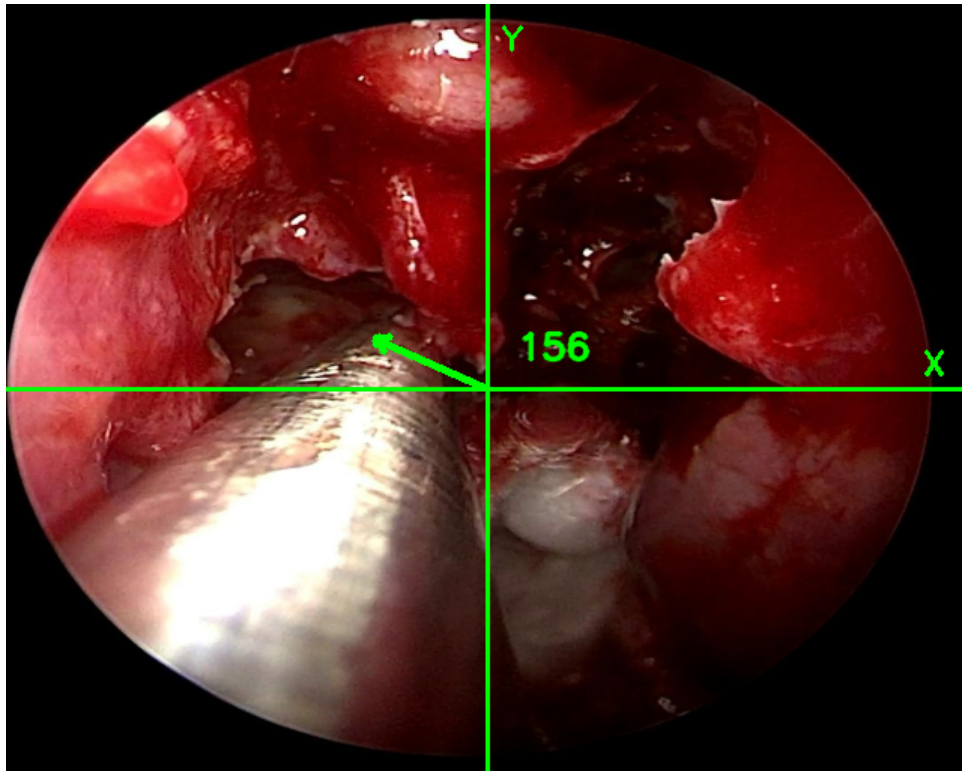


Figure 3.5. Angle calculation for automated guidance of endoscope based on the computed centroid of tool tips.

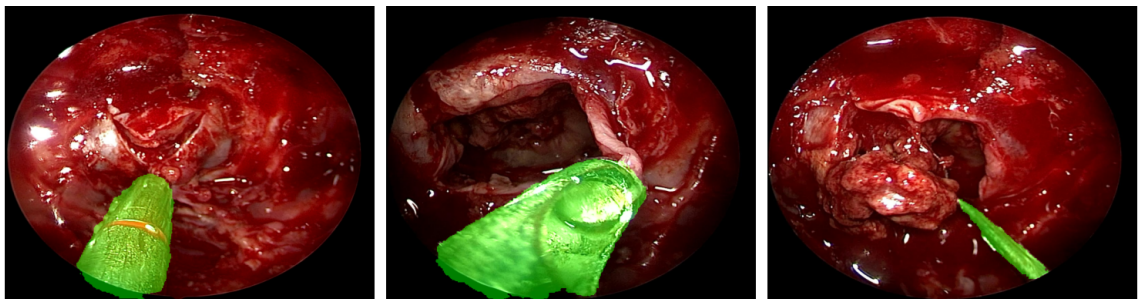


Figure 3.6. From Left to right: Stay Still (7.68% covered), Zoom Out (17.00% covered), and Zoom In (1.66% covered) decision image samples.

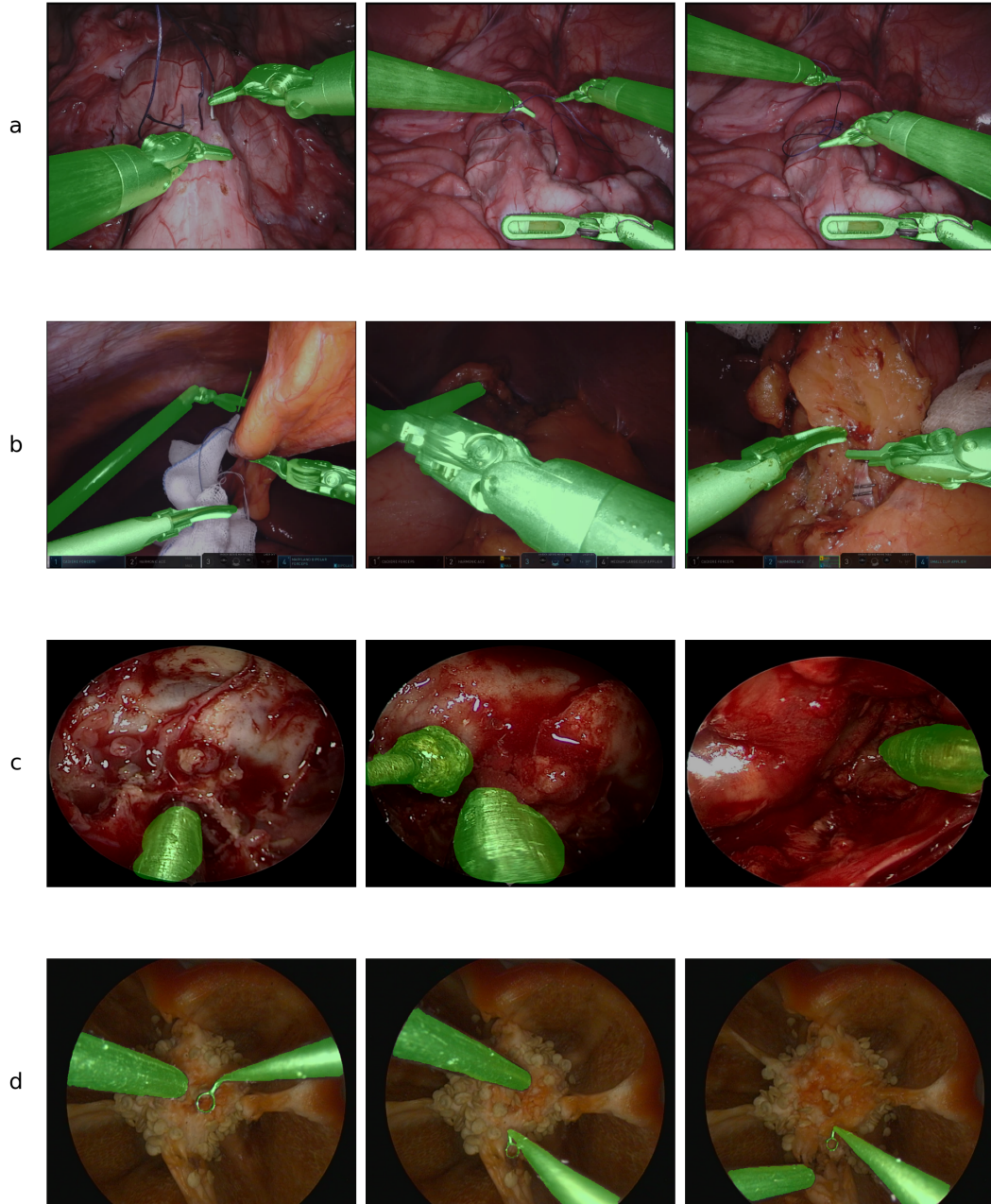


Figure 3.7. Sample images from collected datasets; a) 2017 Endovis Robotic Instrument Segmentation Challenge Test Dataset, b) Surgical Scene Segmentation in Robotic Gastrectomy Dataset, c) Hacettepe University Surgery Dataset, d) Handcrafted Endoscope Dataset .

CHAPTER 4

EXPERIMENTS AND RESULTS

4.1. Training Procedure

The training procedure followed a two-stage approach similar to the methodology adopted in ART-Net (Hasan et al. 2021). In the first stage, only the segmentation network, along with its encoder, was trained. In the second stage, the tip-point sub-network was trained with segmentation sub-network jointly, while the encoder weights were kept frozen to preserve the learned feature representations. The Adagrad optimizer was employed for both stages, with an initial learning rate of 1.0 and an exponential learning rate scheduler with a decay factor of 0.95 was applied for training. The batch size varied between 2 and 8, depending on the available GPU memory, and each stage was trained for a total of 300 epochs. During the first stage training, geometric augmentations such as random horizontal and vertical flips, rotation, and affine scaling, were applied to both images and masks, while image-only augmentations included Gaussian blur and modifications of image brightness, contrast, saturation, and hue.

During validation, the Dice score was used as the evaluation metric for the first stage to select the best-performing checkpoints. The model was validated four times per epoch, and the weights were saved whenever an improvement was observed. For the second stage, joint loss (Equation 4.2), sum of segmentation loss mentioned in Equation 3.2 and MSE Loss for tool tip heatmap training mentioned in Equation 4.1 where N is the number of pixels on image, y_i and $\hat{y}_i$ are predicted heatmap and ground truth label correspondingly, have been employed for both training loss and validation metric to monitor training progress and checkpoint selection. In Equation 4.2 y_{seg} , y_{tt} , $\hat{y}_{seg}$, $\hat{y}_{tt}$ are segmentation ground truth, tool tip ground truth, segmentation prediction, and tool tip heatmap prediction respectively.

Because only approximately 1-2% of the pixels in the ground-truth of tool tip heatmaps are non-zero (Figure 3.4) and the target values are normalized within the range $[0, 1]$, the resulting L_{mse} term has a relatively small numerical magnitude compared to the

segmentation loss L_{seg} . Consequently, its contribution to the overall joint optimization objective becomes negligible. To balance its influence during training, we introduce a scaling factor α applied to L_{mse} . In all experiments, α was empirically set to 5 to ensure that the magnitude of this term is comparable to the other losses in the objective function.

$$L_{mse}(y, \hat{y}) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (4.1)$$

$$L_{joint}(y, \hat{y}) = L_{seg}(y_{seg}, \hat{y}_{seg}) + \alpha L_{mse}(y_{tt}, \hat{y}_{tt}) \quad (4.2)$$

Several loss function combinations were systematically evaluated for the second stage of the training pipeline. These formulations incorporated geometric constraints applied during post-processing. The total loss function was composed of three distinct components: segmentation loss, heatmap loss, and tip validation loss.

The segmentation loss is consistent with that employed in the first training stage—the encoder training phase—as defined in Equation 3.2. The heatmap loss is defined as the mean squared error (MSE) between the predicted heatmap and its corresponding ground truth, as referenced in Equation 4.1. The tip validation loss is introduced to enforce spatial consistency between the predicted activation points and the tool segmentation mask, thereby suppressing false-positive detections, i.e., activated points that do not correspond to any surgical instrument present in the scene. This loss is computed by inverting the segmentation mask such that background regions are assigned a value of one. The predicted heatmap values, normalized to the range $[0, 1)$, are then element-wise multiplied with this inverted mask to isolate activations occurring outside the instrument boundary. The resulting activations are penalized during optimization, encouraging the network to produce tool tip predictions that remain within the segmented instrument regions and improving the spatial reliability of the detected tool locations.

4.2. Evaluation Metrics

A subset of the annotated Hacettepe University Surgery Dataset was used as the test set to evaluate and compare the segmentation performance of the U-Net and ART-

Net architectures. Both models were trained on the same dataset and assessed using the Dice Similarity Coefficient, a standard metric for segmentation quality. The Dice score measures the spatial overlap between a predicted binary mask and its corresponding ground-truth mask, with values ranging from 0 (no overlap) to 1 (perfect overlap). It is defined as twice the cardinality of the intersection between the two masks divided by the sum of their individual cardinalities, as given in Eq. 4.3:

$$Dice(A, B) = \frac{2|A \cap B|}{|A| + |B|} \quad (4.3)$$

where A and B denote the sets of pixels corresponding to the predicted and ground-truth masks, respectively, and $|\cdot|$ represents the set cardinality.

In the second stage of the evaluation, which focuses on predicting instrument tip locations, performance was quantified using the Euclidean distance between the predicted and ground-truth point coordinates in the x–y plane. Because some scenes contain multiple surgical instruments—introducing permutation ambiguity when matching predicted and ground-truth tips—the ground-truth reference was defined as the mean coordinate of all visible tool tips, representing the point at which the endoscope camera should optimally focus.

To complement the distance metric, the number of instruments present in each frame was also estimated. This additional metric enables assessment of not only the accuracy of the predicted focal point but also the model’s ability to distinguish and correctly count the surgical tools within the scene. It is calculated by checking if the tool count matches the ground truth. Only if it matches, the count accuracy is incremented by one.

4.3. Results

4.3.1. Results on Segmentation

In order to compare the ART-Net model and the U-Net model, which was selected as the baseline model for this study, various combinations of mentioned datasets have been collected. The main aim of this experiment is to see how two models generalize when they are evaluated on a test set that has not been present during training. For the

baseline, Surgical Scene Segmentation in Robotic Gastrectomy Dataset has been selected and all experiments included this dataset since it has a nice variety of surgical scenes and includes the largest amount of data compared to other datasets. For other 3 datasets, -the Handcrafted Endoscope Dataset, Hacettepe University Surgical Dataset, and 2017 Endovis Robotic Instrument Segmentation Test Dataset- two of them were combined and added to the training set, while the third one kept for testing. In experiments with models which are trained with all datasets, a custom test set consisting of the Hacettepe dataset and the Handcrafted Endoscope dataset have been created and tested. Due to the scarcity of the Endovis dataset, no image samples from the Endovis dataset were taken. Four trainings were performed with U-Net leaving one dataset out each time, similarly four trainings were performed with ART-Net, and four additional trainings were performed with ART-Net augmented with Convolutional Block Attention Module (CBAM) layers inserted after multiple encoder stages.

The results in Table 4.1 demonstrate varying performance across the three architectures depending on the dataset configuration. U-Net achieved a Dice score of 0.675 when trained with all datasets, indicating moderate overlap between predicted and ground-truth masks. ART-Net obtained a substantially higher Dice score of 0.922 in the same setting, outperforming U-Net by almost 25 percentage points, suggesting that its architectural enhancements and multi-stage design enable more robust feature extraction and more accurate delineation of instrument boundaries. ART-Net with CBAM achieved a comparable score of 0.918 when trained with all datasets, remaining competitive with the base ART-Net.

To further investigate the performance divergence between ART-Net and ART-Net+CBAM, precision and recall metrics were computed for both models trained on the full dataset and evaluated on the custom test set, as reported in Table 4.2. The results reveal a distinct trade-off introduced by the incorporation of CBAM: while the base ART-Net exhibits high recall coupled with comparatively lower precision, the addition of attention layers shifts this balance toward higher precision at the expense of recall. This behavioral shift suggests that the attention mechanism induces greater selectivity in boundary delineation, causing the model to adopt a more conservative prediction strategy around instrument edges. Although such selectivity may confer an advantage in accurately segmenting thicker or more prominently visible instruments, it simultaneously

Table 4.1. Segmentation performance comparison using Dice score

Model	Trained with All Datasets	Excluding Endovis	Excluding Hacettepe	Excluding Handcrafted Endoscope
U-NET	0.675	0.858	0.203	0.285
ART-Net	0.922	0.863	0.586	0.579
ART-Net + CBAM	0.918	0.849	0.654	0.429

increases the likelihood of under-segmentation in cases involving thin instruments or partially occluded tools, where the attended features may be insufficient to confidently activate predictions.

The leave-one-out evaluation reveals that ART-Net consistently outperforms U-Net across all held-out datasets, and ART-Net with CBAM largely sustains or extends this advantage, though with some dataset-dependent variation. While all three models achieve relatively high Dice scores when Endovis is excluded from training (U-Net: 0.858, ART-Net: 0.863, ART-Net+CBAM: 0.849), the differences between models are marginal in this setting, suggesting that the remaining training data provides sufficient coverage and that the models are already near saturation on this domain.

The more informative comparisons arise on the higher-variance configurations. When the Handcrafted Endoscope Dataset is excluded from training and used as the test set, ART-Net (0.579) substantially recovers the severe degradation seen in U-Net (0.285), while ART-Net+CBAM (0.429) falls between the two — a notable regression relative to base ART-Net. This performance drop can be attributed to a key characteristic of the Handcrafted Endoscope Dataset: it predominantly features thinner surgical instruments. As established by the precision-recall analysis, the incorporation of CBAM shifts the model toward higher precision and lower recall, effectively increasing the rate of false negatives. Consequently, when applied to this dataset, ART-Net+CBAM tends to produce incomplete segmentations or entirely miss thin instrument regions, as the attention mechanism enforces overly conservative boundary predictions that are insufficient to capture the full extent of slender tool geometries. This explains the observed Dice score degradation relative to the base ART-Net, which, despite its lower precision, benefits from higher recall and thus

Table 4.2. Segmentation performance comparison using precision and recall

Model	Precision	Recall
ART-Net	0.910	0.953
ART-Net + CBAM	0.955	0.901

Table 4.3. Tool tip localization with morphological pipeline performance comparison

Method	Euclidean Distance (px)	Count Accuracy (%)
U-Net + Morphological	62.441	74.806
ART-Net + Morphological	47.337	90.310
ART-Net + CBAM + Morphological	44.545	87.984

achieves more complete instrument coverage.

When the Hacettepe dataset is excluded, which consistently emerges as the most challenging configuration, U-Net suffers a severe performance drop (0.203), while both ART-Net variants maintain considerably more stable segmentation accuracy — ART-Net achieving 0.586 and ART-Net+CBAM improving upon it at 0.654. This performance gap can be partly attributed to the distinctive characteristics of the Hacettepe dataset, which predominantly contains scenes with a single, thick surgical instrument. In such a setting, overall segmentation accuracy is less dependent on detecting multiple or partially visible tools, and is instead primarily determined by the precision of boundary delineation along the instrument’s edges. As established by the precision-recall analysis, ART-Net+CBAM’s attention mechanism induces greater selectivity in edge prediction, which proves particularly advantageous in this context: the model’s conservative, high-precision boundary predictions align well with the clearly defined and spatially dominant instrument contours present in this dataset. This explains why ART-Net+CBAM achieves the highest Dice score in this configuration, as its tendency to prioritize precise edge localization over broad coverage becomes a strength rather than a limitation when the target instrument is large, singular, and well-defined.

Table 4.4. Tool tip localization from heatmap performance comparison

Method	Euclidean Distance (px)	Count Accuracy (%)
ART-Net Heatmap	32.155	83.720
ART-Net Heatmap w/ Head-only CBAM	35.685	81.782
ART-Net Heatmap w/ CBAM	62.832	73.643

4.3.2. Results on Tool Tip Localization

The second stage results are evaluated constructing a test dataset that includes the Hand-crafted Endoscope Dataset and Hacettepe University Surgical Dataset. Two different algorithms, morphological and deep learning based tool tip detection, were tested on this dataset and results were collected. The morphological algorithm has applied both on U-Net, ART-Net, and ART-Net with CBAM segmentation output. The performance of the tool tip detection component was evaluated using two complementary metrics: Euclidean distance and count accuracy. Euclidean distance measures the pixel-wise distance between the predicted focal point—defined as the average of the detected tool tip coordinates—and the ground-truth point representing the ideal camera focus location. Lower values indicate more precise localization.

The morphological post-processing pipeline results, presented in Table 4.3, demonstrate a clear and consistent performance hierarchy across all three architectural configurations. ART-Net and ART-Net+CBAM substantially outperform U-Net in both metrics, with U-Net yielding the highest Euclidean distance (62.441 px) and the lowest count accuracy (74.806%). Among the ART-Net variants, ART-Net+CBAM achieves the most precise localization in terms of Euclidean distance (44.545 px), while base ART-Net attains the highest count accuracy (90.310%), with ART-Net+CBAM remaining competitive at 87.984%. This marginal divergence between the two ART-Net variants is consistent with the precision-recall trade-off introduced by CBAM integration: the more conservative boundary predictions produced by the attention mechanism yield spatially tighter segmentation masks that facilitate more precise tip coordinate estimation, yet the elevated false negative rate marginally reduces the model’s ability to detect all instruments present in the scene, resulting in a slight drop in count accuracy. Overall, the morphological pipeline

results confirm that segmentation quality directly and positively influences downstream localization performance, with ART-Net-based architectures providing a substantially more reliable foundation for post-processing-based tip detection than U-Net.

The heatmap-based localization results, presented in Table 4.4, reveal a consistent degradation in performance as CBAM integration is progressively broadened across the network. The base ART-Net heatmap model achieves the strongest performance across both metrics, attaining a Euclidean distance of 32.155 px and a count accuracy of 83.720%. Incorporating CBAM exclusively within the heatmap prediction head results in a moderate reduction in localization precision (35.685 px) and count accuracy (81.782%), while extending CBAM across the full network leads to a substantially more pronounced degradation, with Euclidean distance increasing to 62.832 px and count accuracy dropping to 73.643%. This progressive deterioration with broader attention integration suggests a fundamental incompatibility between the spatial feature suppression enforced by CBAM and the requirements of precise keypoint localization. Unlike segmentation, which benefits from selective boundary emphasis, heatmap regression demands the preservation of fine-grained and spatially distributed activations in order to accurately resolve the precise location of instrument tips. The broader the scope of CBAM integration, the more aggressively these activations are suppressed, ultimately compromising the network’s localization capability.

Comparing the two localization paradigms, the heatmap-based approach demonstrates a clear advantage in Euclidean distance, with the best-performing heatmap model (32.155 px) reducing localization error by approximately 12 px relative to the best morphological approach (44.545 px), corresponding to a reduction of roughly 28%. However, the morphological pipeline achieves a notably higher count accuracy in its best configuration (ART-Net+Morphological, 90.310%) compared to the best heatmap model (83.720%), suggesting that segmentation-driven approaches retain a meaningful advantage in instrument enumeration tasks. This complementary performance profile indicates that the two approaches capture different aspects of the localization problem: the heatmap regression framework excels at precise spatial localization by learning direct feature-to-coordinate mappings, while the morphological pipeline leverages the structural integrity of the segmentation mask to more reliably enumerate distinct instrument instances. These findings suggest that a hybrid strategy combining heatmap-based coordinate regression with

segmentation-driven instrument counting could potentially yield superior performance across both metrics.

4.3.3. Results on Tool Presence Accuracy

While the experiments presented thus far were conducted exclusively on scenes containing at least one surgical instrument, real surgical footage may also include scenes in which no instruments are present. To evaluate the models' capability to distinguish between tool-present and tool-absent scenes, an additional experiment was conducted using images collected from the Handcrafted Endoscope Dataset. This experiment comprised two scene categories, tool-present and tool-absent, with 100 images per class, yielding a total of 200 images. The evaluation was restricted to the two tool tip localization pipelines: the morphological pipeline and the heatmap-based detector. An image was classified as tool-present if the pipeline detected at least one tool tip, and as tool-absent otherwise.

For the morphological pipeline, tool presence classification is dependent on the underlying segmentation output. If the segmentation model produces false-positive activations that are not filtered out during post-processing, the pipeline proceeds to apply morphological post-processing and reports a tool tip, thereby misclassifying the scene as tool-present. The heatmap-based detector instead determines tool presence based on the predicted heatmap activations: if activation values exceed a defined threshold and the corresponding candidate tip locations are consistent with the segmentation output, the instrument is classified as present; otherwise, the scene is classified as tool-absent.

A total of six tool tip detection configurations were evaluated, with results reported in Table 4.5. The base ART-Net model achieved perfect classification accuracy (100.0%) under both pipelines, while U-Net performed at chance level (50.0%), indicating an inability to reliably discriminate tool-present from tool-absent scenes. CBAM integration in the backbone substantially degraded performance (61.0% and 50.5%), whereas restricting CBAM to the head preserved near-perfect accuracy (98.5%)—consistent with the earlier finding that frozen, segmentation-biased backbone activations undermine task-specific discrimination.

It is important to contextualize these classification results within the intended operational setting of the system. In the simulated surgical environment, the endoscope

Table 4.5. Tool presence/absence classification comparison

Method	Classification Accuracy (%)
U-Net Morphological	50.0
ART-Net Morphological	100.0
ART-Net + CBAM Morphological	61.0
ART-Net Heatmap	100.0
ART-Net Heatmap w/ Head-only CBAM	98.5
ART-Net Heatmap w/ CBAM	50.5

repositioning is only triggered upon the surgeon’s deliberate activation via a foot pedal. Consequently, false positives or missed detections do not impose any direct burden on the surgical workflow — in the absence of an active tool or in moments where the surgeon is otherwise occupied, no pedal input is provided and the system remains idle. This surgeon-in-the-loop design effectively mitigates the practical impact of classification errors, suggesting that even configurations with suboptimal tool presence accuracy remain viable within the context of the intended use case.

CHAPTER 5

CONCLUSION

This thesis presented a comprehensive investigation into deep learning-based surgical instrument segmentation and tool tip localization, with a particular focus on evaluating the generalization capability of encoder-decoder architectures across heterogeneous endoscopic datasets. Three architectural configurations were systematically compared — U-Net as the baseline, ART-Net as the primary model, and ART-Net augmented with Convolutional Block Attention Module (CBAM) layers — under a rigorous leave-one-out experimental protocol designed to assess cross-domain robustness rather than in-distribution performance alone.

The segmentation experiments demonstrated that ART-Net substantially and consistently outperforms U-Net across all dataset configurations, confirming that its multi-stage architectural design and enhanced feature extraction capability confer significant advantages in the challenging and visually diverse domain of surgical endoscopy. The integration of CBAM introduced a dataset-dependent trade-off: precision-recall analysis revealed that attention mechanisms systematically shift the model toward higher precision and lower recall, producing more conservative boundary predictions. This property proved beneficial in scenarios involving thick, spatially dominant instruments with well-defined edges, as evidenced by the superior performance on the Hacettepe Surgical Dataset, but manifested as a liability on datasets featuring thin or partially occluded instruments, where the elevated false negative rate led to incomplete segmentations and reduced Dice scores.

The second stage of the pipeline extended the segmentation backbone toward tool tip localization through two complementary approaches: a morphological post-processing pipeline operating on binary segmentation masks, and a jointly trained dual-head network combining segmentation with heatmap-based tip regression. The morphological pipeline demonstrated that localization performance largely mirrors segmentation quality, with ART-Net and ART-Net+CBAM substantially outperforming U-Net in both Euclidean distance and count accuracy. Notably, ART-Net+CBAM achieved the most precise spatial localization while base ART-Net retained a slight advantage in count accuracy, a divergence that is consistent with the precision-recall trade-off characterizing CBAM integration —

tighter boundary predictions facilitate more accurate tip coordinate estimation, while the elevated false negative rate marginally reduces instrument enumeration reliability. The heatmap-based approach achieved superior localization precision overall, with the base ART-Net heatmap model attaining the lowest Euclidean distance across both paradigms, demonstrating that direct feature-to-coordinate regression bypasses the dependency on segmentation mask integrity and yields more precise spatial estimates. However, the morphological pipeline retained a meaningful advantage in count accuracy, suggesting that segmentation-driven approaches are more reliable for instrument enumeration. Notably, CBAM integration proved consistently detrimental in the heatmap localization task, with performance degrading progressively as attention was applied more broadly across the network, confirming that the selective spatial suppression enforced by attention mechanisms fundamentally conflicts with the fine-grained spatial retention required for precise keypoint localization.

Collectively, the findings of this thesis highlight several important insights for the design of surgical instrument perception systems. First, architectural complexity alone is insufficient to guarantee generalization; training set diversity and domain coverage play an equally critical role in cross-domain robustness. Second, attention mechanisms such as CBAM do not universally improve performance and their benefit is inherently conditioned on the geometric and visual properties of the target domain. Third, direct heatmap regression offers a compelling and robust alternative to purely segmentation-driven localization pipelines, particularly in scenarios where segmentation quality is compromised by domain shift or instrument geometry.

Future work may explore adaptive attention mechanisms that dynamically modulate their selectivity based on estimated instrument geometry, as well as more principled strategies for domain adaptation to enhance cross-domain generalization across diverse surgical tool categories. Beyond binary segmentation masks, the incorporation of keypoint detection as a complementary representational modality could provide richer structural descriptors of instrument morphology and pose. Furthermore, the integration of temporal modeling — through recurrent architectures or optical flow-based approaches — may enable robust multi-frame instrument tracking, which is critical for continuous intraoperative monitoring. Finally, extending the proposed pipeline to support real-time inference would constitute a meaningful and necessary step toward clinical deployment.

REFERENCES

- Allan, M., S. Ourselin, D. J. Hawkes, J. D. Kelly, and D. Stoyanov. 2018. “3-D Pose Estimation of Articulated Instruments in Robotic Minimally Invasive Surgery.” *IEEE Transactions on Medical Imaging* 37 (5): 1204–1213. <https://doi.org/10.1109/TMI.2018.2794439>.
- Allan, Max, Alexey A. Shvets, Thomas Kurmann, Zichen Zhang, Rahul Duggal, Yun-Hsuan Su, Nicola Rieke, et al. 2019. “2017 Robotic Instrument Segmentation Challenge.” *ArXiv* abs/1902.06426. <https://api.semanticscholar.org/CorpusID:67787647>.
- Bouarfa, Loubna, Oytun Akman, Armin Schneider, Pieter P. Jonker, and Jenny Dankelman. 2012. “In-vivo Real-Time Tracking of Surgical Instruments in Endoscopic Video.” *Minimally Invasive Therapy & Allied Technologies* 21 (3): 129–134. <https://doi.org/10.3109/13645706.2011.580764>.
- Bouget, David, Max Allan, Danail Stoyanov, and Pierre Jannin. 2016. “Vision-Based and Marker-Less Surgical Tool Detection and Tracking: a Review of the Literature.” *Medical Image Analysis* 35 (September). <https://doi.org/10.1016/j.media.2016.09.003>.
- Cerón, Juan Carlos Ángeles, Gilberto Ochoa Ruiz, Leonardo Chang, and Sharib Ali. 2022. “Real-time Instance Segmentation of Surgical Instruments Using Attention and Multi-Scale Feature Fusion.” *Medical Image Analysis* 81:102569. <https://doi.org/10.1016/j.media.2022.102569>.
- Chakradeo, Vinaya, Kavita Malhotra, Horacio D’Agostino, and Ravish Patwardhan. 2006. “The Voice-Controlled Robotic Assist Scope Holder AESOP for the Endoscopic Approach to the Sella.” *Skull Base* 16 (May): 123–131. <https://doi.org/10.1055/s-2006-939679>.
- Chen, Jieneng, Jieru Mei, Xianhang Li, Yongyi Lu, Qihang Yu, Qingyue Wei, Xiangde Luo, et al. 2024. “TransUNet: Rethinking the U-Net Architecture Design for Medical Image Segmentation Through the Lens of Transformers.” *Medical Image Analysis* 97 (October): 103280. <https://doi.org/10.1016/j.media.2024.103280>.

- Chmarra, M. K., C. A. Grimbergen, and J. Dankelman. 2007. "Systems for tracking minimally invasive surgical instruments." *Minimally Invasive Therapy & Allied Technologies* 16 (6): 328–340. <https://doi.org/10.1080/13645700701702135>.
- Colleoni, Emanuele, Philip Edwards, and Danail Stoyanov. 2020. "Synthetic and Real Inputs for Tool Segmentation in Robotic Surgery." In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2020*, edited by Anne L. Martel, Purang Abolmaesumi, Danail Stoyanov, Diana Mateus, Maria A. Zuluaga, S. Kevin Zhou, Daniel Racoceanu, and Leo Joskowicz, 700–710. Cham: Springer International Publishing.
- Colleoni, Emanuele, and Danail Stoyanov. 2021. "Robotic Instrument Segmentation With Image-to-Image Translation." *IEEE Robotics and Automation Letters* 6 (2): 935–942. <https://doi.org/10.1109/LRA.2021.3056354>.
- Dede, Can, Gökhan Kiper, Yalın Baştanlar, Şenay Mihçin, Omar Maarooft, Mustafa Berker, İlkey Işıkkay, and Şahin Hanalioglu. 2023. *CleverScope: The Control of the Endoscope Router Robot to be used in Endonasal Surgeries Supported by Deep Learning Algorithms*. Iztech Robotics Lab, <https://robotics.iyte.edu.tr/cleverscope/>. TÜBİTAK Project No: 123M353. Accessed: June 2026.
- Dede, Mehmet, Gökhan Kiper, Tolga Ayav, Barbaros Ozdemirel, Enver Tatlicioglu, Sahin Hanalioglu, Ilkay Isikay, and Mustafa Berker. 2020. "Human-Robot Interfaces of the NeuRoboScope: A Minimally Invasive Endoscopic Pituitary Tumor Surgery Robotic Assistance System." *Journal of Medical Devices* 15 (December). <https://doi.org/10.1115/1.4049394>.
- Deininger, L., S. Jung-Klawitter, and R. Mikut. 2023. "An AI-based Segmentation and Analysis Pipeline for High-Field MR Monitoring of Cerebral Organoids." *Scientific Reports* 13:21231. <https://doi.org/10.1038/s41598-023-48343-7>.

- Du, Xiaofei, Thomas Kurmann, Ping-Lin Chang, Maximilian Allan, Sebastien Ourselin, Raphael Sznitman, John D. Kelly, and Danail Stoyanov. 2018. "Articulated Multi-Instrument 2-D Pose Estimation Using Fully Convolutional Networks." *IEEE Transactions on Medical Imaging* 37 (5): 1276–1287. <https://doi.org/10.1109/TMI.2017.2787672>.
- Fujii, Kenko, Gauthier Gras, Antonino Salerno, and Guang-Zhong Yang. 2017. "Gaze Gesture Based Human Robot Interaction for Laparoscopic Surgery." *Medical Image Analysis* 44 (November). <https://doi.org/10.1016/j.media.2017.11.011>.
- Gruijthuijsen, Caspar, Luis C. Garcia-Peraza-Herrera, Gianni Borghesan, Dominiek Reynaerts, Jan Deprest, Sebastien Ourselin, Tom Vercauteren, and Emmanuel Vander Poorten. 2022. "Robotic Endoscope Control Via Autonomous Instrument Tracking." *Frontiers in Robotics and AI* 9 (April). <https://doi.org/10.3389/frobt.2022.832208>.
- Hasan, Md. Kamrul, Lilian Calvet, Navid Rabbani, and Adrien Bartoli. 2021. "Detection, Segmentation, and 3D Pose Estimation of Surgical Tools Using Convolutional Neural Networks and Algebraic Geometry." *Medical Image Analysis* 70:101994. <https://doi.org/10.1016/j.media.2021.101994>.
- Iovene, Elisa, Diego Cattaneo, Junling Fu, Giancarlo Ferrigno, and Elena De Momi. 2024. "Hybrid Tracking Module for Real-Time Tool Tracking for an Autonomous Exoscope." *IEEE Robotics and Automation Letters* 9 (7): 6067–6074. <https://doi.org/10.1109/LRA.2024.3400124>.
- Kahveci, Burak, Elifsu Polatli, Yalin Bastanlar, and Sinan Guven. 2024. "OrganoLabeler: A Quick and Accurate Annotation Tool for Organoid Images." *ACS Omega* 9, no. 46 (November): 46117–46128. <https://doi.org/10.1021/acsomega.4c06450>.
- Kahveci, Burak, Elifsu Polatli, Ali Eren Evranos, Hüseyin Güner, Gökhan Karakulah, Yalin Bastanlar, and Sinan Güven. 2026. "BrAIIn: A Comprehensive Artificial Intelligence-Based Morphology Analysis System for Brain Organoids and Neuroscience." *Bio-engineering & Translational Medicine* 11 (3): e70123. <https://doi.org/10.1002/btm2.70123>.

- Kanakatte, Aparna, Anitha Ramaswamy, Jayavardhana Gubbi, Avik Ghose, and Balamuralidhar Purushothaman. 2020. "Surgical Tool Segmentation and Localization Using Spatio-Temporal Deep Network." *Annual International Conference of the IEEE Engineering in Medicine and Biology Society* 2020:1658–1661. <https://doi.org/10.1109/EMBC44109.2020.9176676>.
- Kim, Jun, Won Cheol Park, and Joo Lee. 2019. "Comparison of Short-term Outcomes of Laparoscopic-Assisted Colon Cancer Surgery Using a Joystick-Guided Endoscope Holder (Soloassist II) or a Human Assistant." *Annals of Coloproctology* 35 (August): 181–186. <https://doi.org/10.3393/ac.2018.10.18>.
- Ko, Seong-Young, Jonathan Kim, Dong-Soo Kwon, and Woo-Jung Lee. 2005. "Intelligent Interaction Between Surgeon and Laparoscopic Assistant Robot System." *Proceedings - IEEE International Workshop on Robot and Human Interactive Communication* 2005:60–65. <https://doi.org/10.1109/ROMAN.2005.1513757>.
- Laina, Iro, Nicola Rieke, Christian Rupprecht, Josué Page Vizcaíno, Abouzar Eslami, Federico Tombari, and Nassir Navab. 2017. "Concurrent Segmentation and Localization for Tracking of Surgical Instruments." *Medical Image Computing and Computer-Assisted Intervention* (Cham), 664–672.
- Lee, Dongheon, Hyeong Won Yu, Hyungju Kwon, Hyoun-Joong Kong, Kyu Eun Lee, and Hee Chan Kim. 2020. "Evaluation of Surgical Skills during Robotic Surgery by Deep Learning-Based Multiple Surgical Instrument Tracking in Training and Actual Operations." *Journal of Clinical Medicine* 9 (6). <https://doi.org/10.3390/jcm9061964>.
- Pfeiffer, Micha, Isabel Funke, Maria R. Robu, Sebastian Bodenstedt, Leon Strenger, Sandy Engelhardt, Tobias Roß, et al. 2019. "Generating Large Labeled Data Sets for Laparoscopic Image Processing Tasks Using Unpaired Image-to-Image Translation." In *Medical Image Computing and Computer Assisted Intervention - MICCAI 2019 - 22nd International Conference, Shenzhen, China, October 13-17, 2019, Proceedings, Part V*, edited by Dinggang Shen, Tianming Liu, Terry M. Peters, Lawrence H. Staib, Caroline Essert, Sean Zhou, Pew-Thian Yap, and Ali R. Khan, 11768:119–127.

- Lecture Notes in Computer Science. Springer. https://doi.org/10.1007/978-3-030-32254-0_14.
- Ronneberger, Olaf, Philipp Fischer, and Thomas Brox. 2015. “U-Net: Convolutional Networks for Biomedical Image Segmentation.” In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, edited by Nassir Navab, Joachim Hornegger, William M. Wells, and Alejandro F. Frangi, 234–241. Cham: Springer International Publishing.
- Sahu, Manish, Ronja Strömsdörfer, Anirban Mukhopadhyay, and Stefan Zachow. 2020. “Endo-Sim2Real: Consistency Learning-Based Domain Adaptation for Instrument Segmentation.” In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2020*, edited by Anne L. Martel, Purang Abolmaesumi, Danail Stoyanov, Diana Mateus, Maria A. Zuluaga, S. Kevin Zhou, Daniel Racoceanu, and Leo Joskowicz, 784–794. Cham: Springer International Publishing.
- Shen, Wenting, Yaonan Wang, Min Liu, Jiazheng Wang, and Renjie Ding. 2026. “UniSurg: A Unified Multitask Framework for Robotic Surgical Scene Understanding.” *IEEE Transactions on Circuits and Systems for Video Technology* 36 (5): 6184–6196. <https://doi.org/10.1109/TCSVT.2026.3651960>.
- Speidel, Stefanie, Michael Delles, Carsten Gutt, and Rüdiger Dillmann. 2006. “Tracking of Instruments in Minimally Invasive Surgery for Surgical Skill Analysis.” In *Medical Imaging and Augmented Reality*, edited by Guang-Zhong Yang, TianZi Jiang, Dinggang Shen, Lixu Gu, and Jie Yang, 148–155. Berlin, Heidelberg: Springer Berlin Heidelberg.
- Stott, Martyn C., Jenifer Barrie, Domnic Sebastien, Christa Hammill, and Daren A. Subar. 2017. “Is the Use of a Robotic Camera Holder Economically Viable? A Cost Comparison of Surgical Assistant Versus the Use of a Robotic Camera Holder in Laparoscopic Liver Resections.” *Surgical Laparoscopy, Endoscopy & Percutaneous Techniques* 27, no. 5 (October): 375–378. <https://doi.org/10.1097/SLE.0000000000000452>.

- Tonet, Oliver, Ramesh U. Thoranaghatte, Giuseppe Megali, and Paolo Dario. 2007. "Tracking Endoscopic Instruments Without a Localizer: A Shape-Analysis-Based Approach." *Computer Aided Surgery* 12 (1): 35–42. <https://doi.org/10.1080/10929080701210782>.
- Woo, Sanghyun, Jongchan Park, Joon-Young Lee, and In So Kweon. 2018. "CBAM: Convolutional Block Attention Module." In *Computer Vision – ECCV 2018: 15th European Conference, Munich, Germany, September 8–14, 2018, Proceedings, Part VII*, 3–19. Munich, Germany: Springer-Verlag. https://doi.org/10.1007/978-3-030-01234-2_1.
- Yoon, Jihun, SeulGi Hong, Seungbum Hong, Jiwon Lee, Soyeon Shin, Bokyoung Park, Nakjun Sung, et al. 2022. "Surgical Scene Segmentation Using Semantic Image Synthesis with a Virtual Surgery Environment." In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2022*, edited by Linwei Wang, Qi Dou, P. Thomas Fletcher, Stefanie Speidel, and Shuo Li, 551–561. Cham: Springer Nature Switzerland.
- Zhang, Chi, Wangru Zhu, Jianqing Peng, Yu Han, and Wanquan Liu. 2023. "Visual Servo Control of Endoscope-Holding Robot Based on Multi-Objective Optimization: System Modeling and Instrument Tracking." *Measurement* 211:112658. <https://doi.org/10.1016/j.measurement.2023.112658>.
- Zhang, Youqiang, Minhyo Kim, Jun Seok Park, and Sangrok Jin. 2026. "Consistent Identification of Laparoscopic Tools by Direction and Angle-Guided Tracking for Robot-Assisted Surgery." *IEEE Robotics and Automation Letters* 11 (2): 1722–1729. <https://doi.org/10.1109/LRA.2025.3643334>.